

Modelo de Componentes No Observados aplicado al gasto en publicidad de una institución bancaria.

Caso práctico aplicado al análisis de series de tiempo

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1. Trayectoria profesional

2001: Egresado de Acturía en UNAM Acatlán

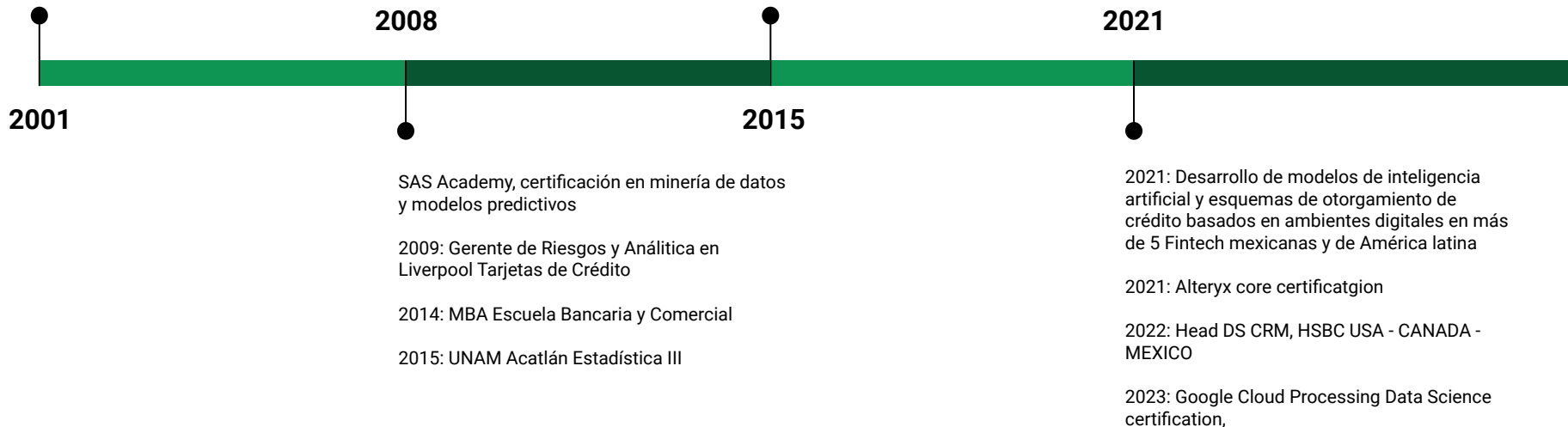
2004: Paris-Francia Responsable de sistemas de ayuda a la decisión en BNP Paribas

2008: Gerente Sr. Riesgo y Modelos Inxev Tarjetas de Crédito

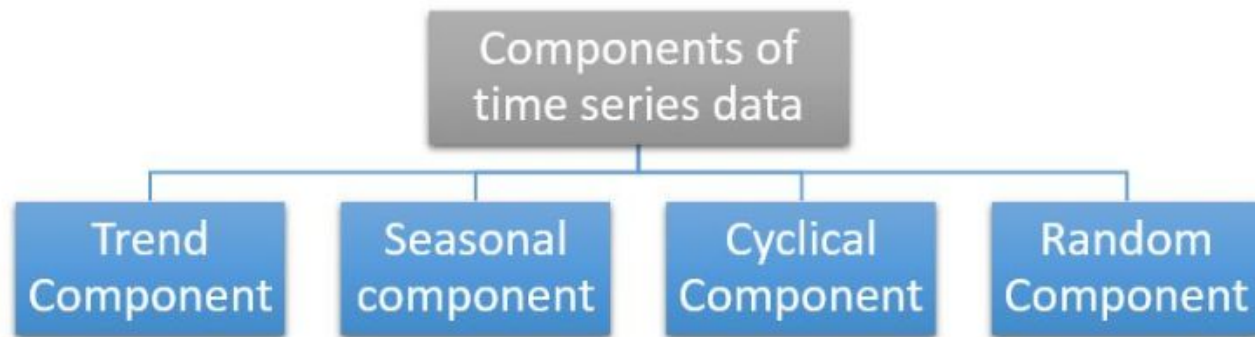
2016: SAS Academy, certificación en Machine Learning e Inteligencia Artificial (USA)

2018: Director de Riesgo de Crédito y Modelos en Banco Azteca

2020: Maestría en Ciencia de Datos y Big Data, Universidad de la Rioja



2. Basic concepts



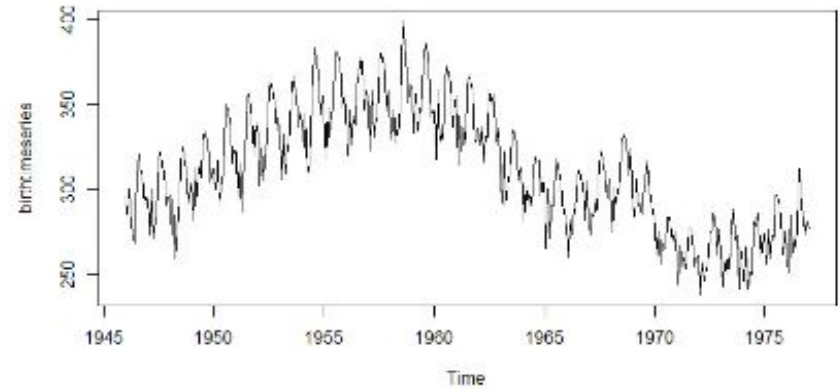
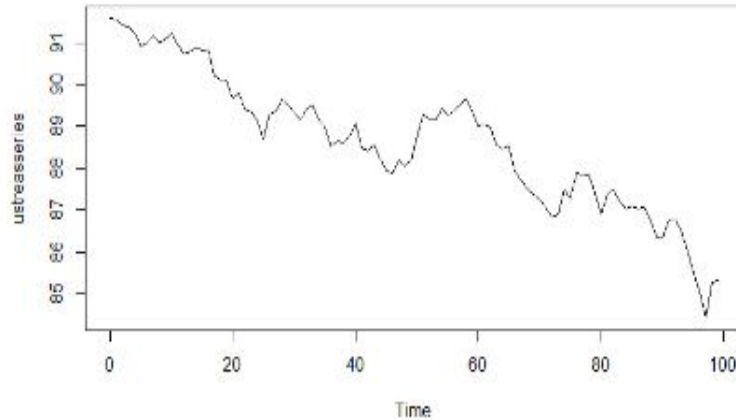
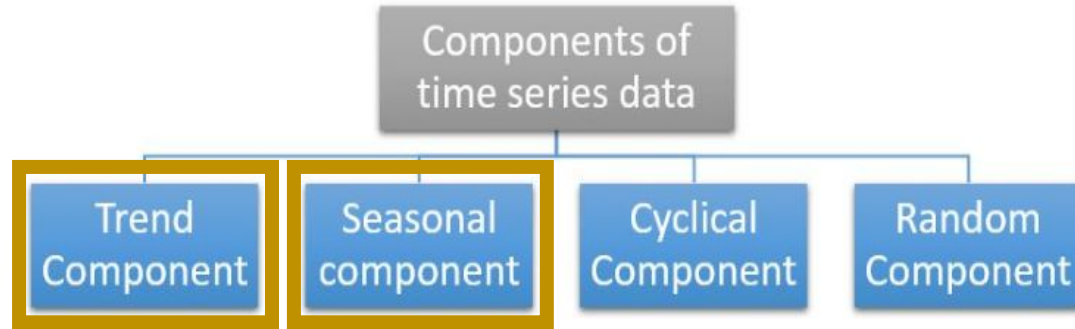
Trend Component: By trend component, we mean that the general tendency of the data to increase or decrease during a long period of time.

Seasonal Component: The variations in the time series that arise due to the rhythmic forces which operate over a span of less than 12 months or a year.

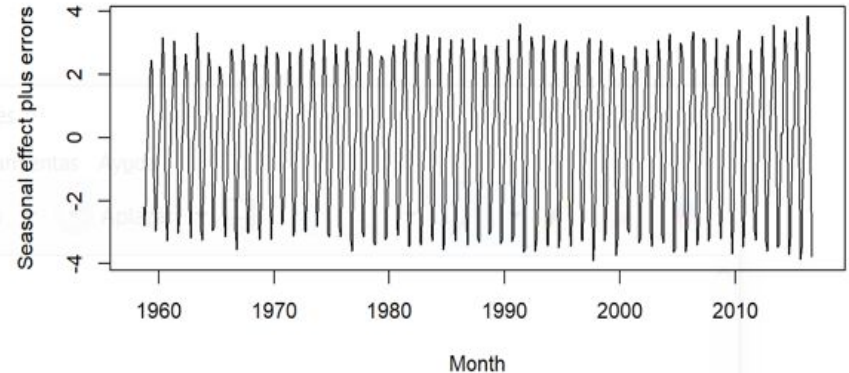
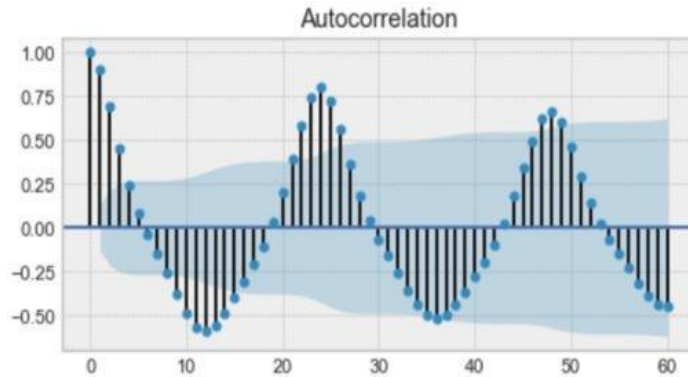
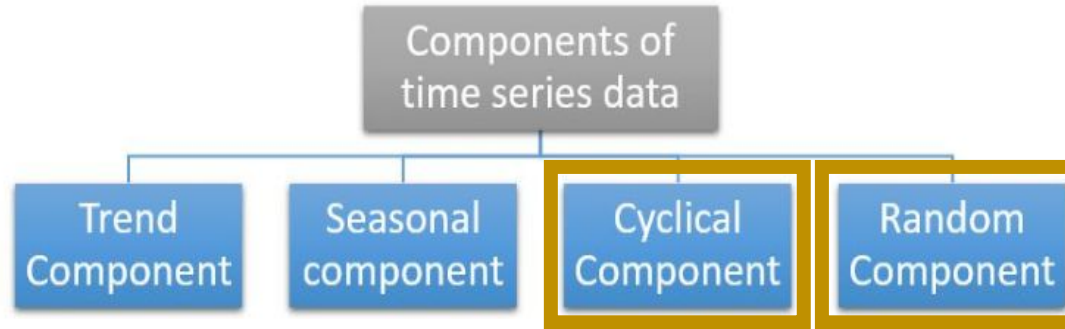
Cyclical Component: The oscillatory movements in a time series that last for more than a year.

Random Component: Random or irregular variations or fluctuations which are not accounted for by trend, seasonality and cyclical components are defined as the random component. These are also called episodic fluctuations.

2. Basic concepts

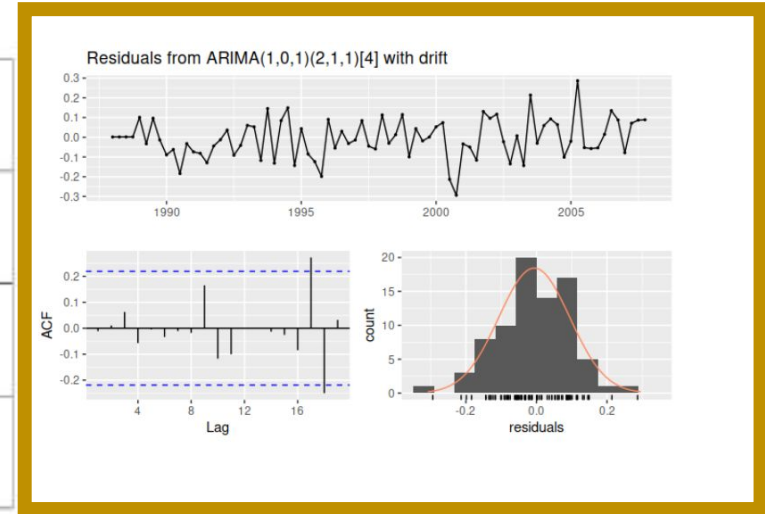
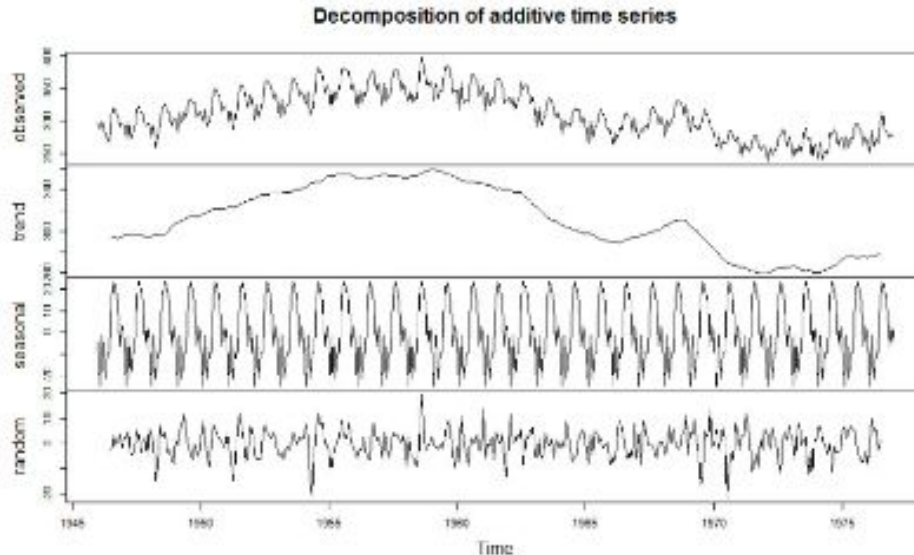


2. Basic concepts



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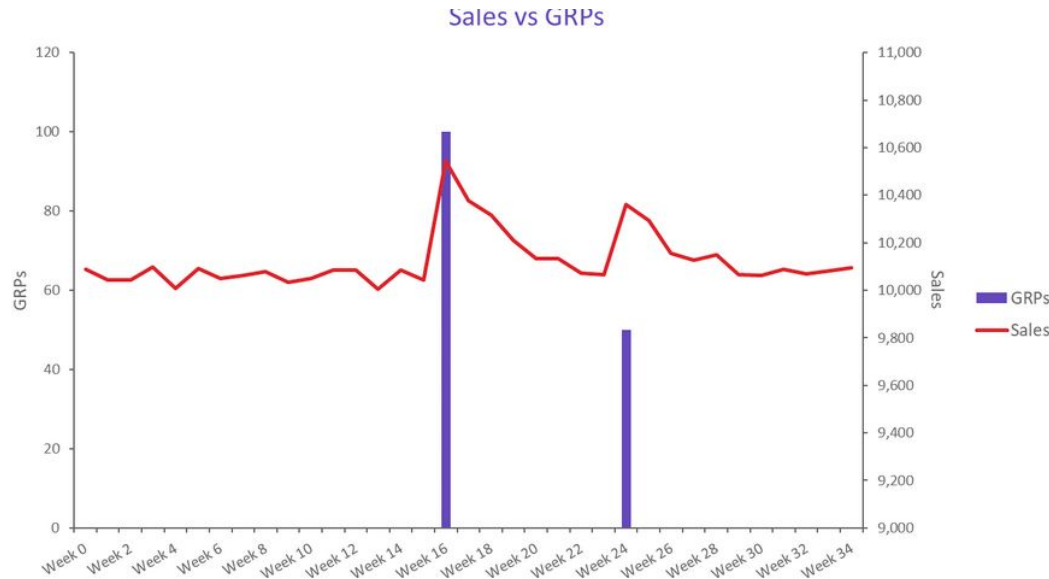
UCM Model¹: Unobserved Components Model (Harvey (1989)) performs a time series decomposition into components such as trend, seasonal, cycle, and the regression effects due to predictor series



¹ <https://medium.com/analytics-vidhya/tsa-ucm-python-5fde69d42e28>

2. Basic concepts

Adstock² (sometimes referred to as “carryover” or “decay”) is a concept first outlined by Simon Broadbent in 1979. It refers to the persistence of an advertising effect beyond the period in which the message was delivered.

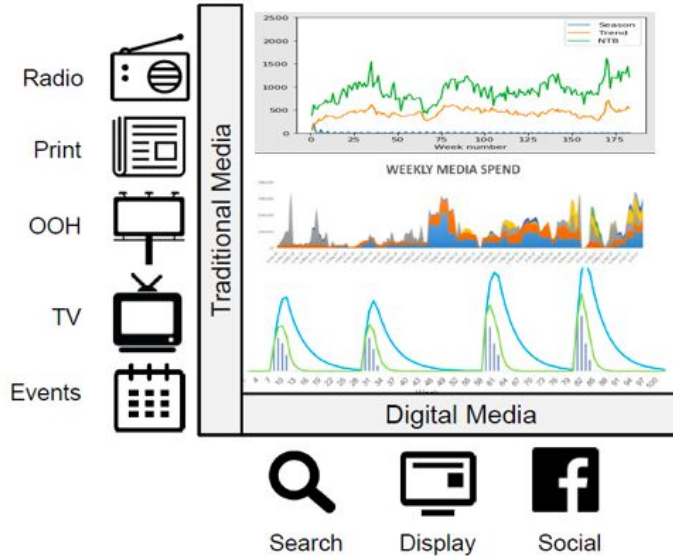


Usually this can be seen in a sales chart as a tailing off of the incremental sales following the period of advertising. The below chart shows GRPs (a TV-campaign specific metric) plotted against sales. Note how each spike in TV ad spend seems to cause a period of elevated sales for some weeks after.

² <https://getrecast.com/adstock-rates/>

3. Media Mix Model (MMM)

- Media Mix Model is an attribution to New To Bank (NTB) acquisition to marketing spend
- The model is used to predict the NTB projection within next 12 months
- In the future cookieless world MMM / attribution models will play a key role in business optimization



Weekly NTB
(Jan 2019 – Dec 2022)

Media Mix Modelling is a methodology for **attributing sales to Marketing** spend by media.

This attribution has two purposes:

1. Measure the efficiency of historical Marketing spend
2. Optimize future Marketing spend

3. Media Mix Model (MMM)

What is included for this MMM exercise



H2'22 data added

- previous version of the model covered Jan'19-Jun'22 period
- current version of the models covers Jan'19-Dec'22 period



Affiliates & Events media channels added

- As the media channel was activated and spend on the channel increased, it was critical to consider the channel in the model
- Spend on Events was also included in the model, but expectations re: statistical significance of the channel and consistent lag/decay between spend on events and NTB are low due to complexity of impact of events



Google Trends data added

- Through experimentation, the following term search terms were identified that correlated the most with the NTB: "HSBC Credit Card" and "HSBC Savings Account"
- Only search terms originated from Canada were considered for the model

3. Media Mix Model (MMM)

Input data

Marketing Spend (Jan'19 – Dec'22)				
Media		Spend \$	Spend %	Coverage %
TV			2%	29%
Display			16%	99%
Search			26%	100%
Newspaper			3%	37%
Outdoor			7%	45%
Radio			13%	62%
Social	   		8%	94%
Magazine			1%	28%
Programmatic			5%	38%
Affiliates			11%	75%
Events			6%	12%

5

Macroeconomic and Seasonal Data*

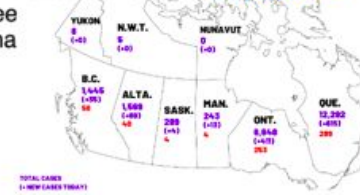


Immigration, Refugee and Citizenship Cana



Statistics Canada

CASES OF COVID-19 IN CANADA

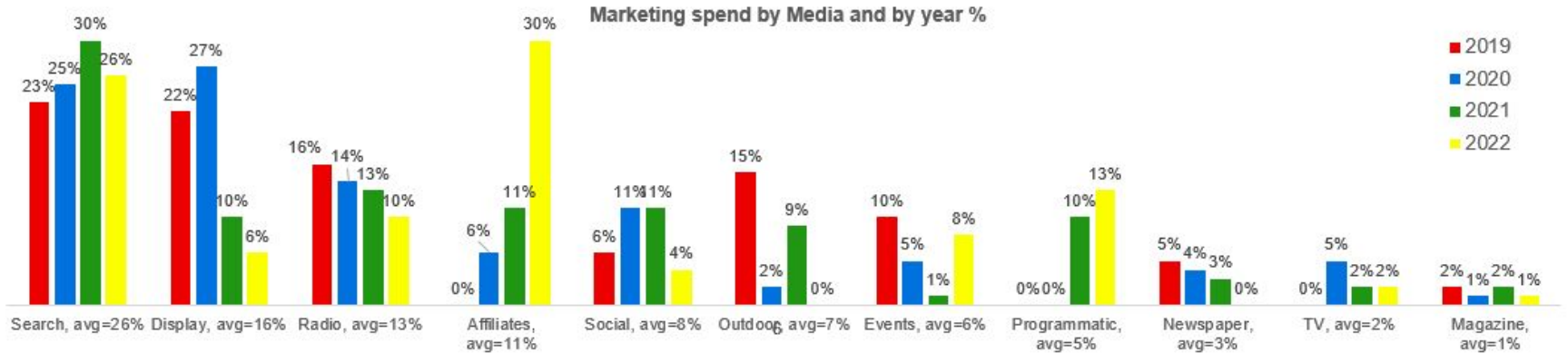
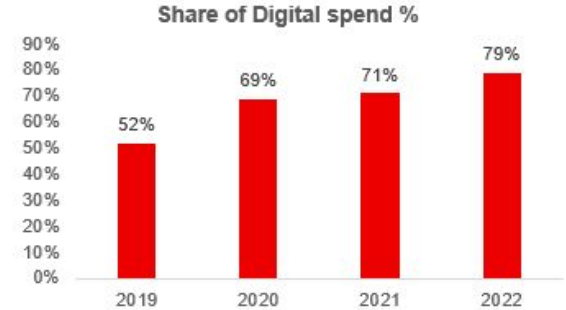
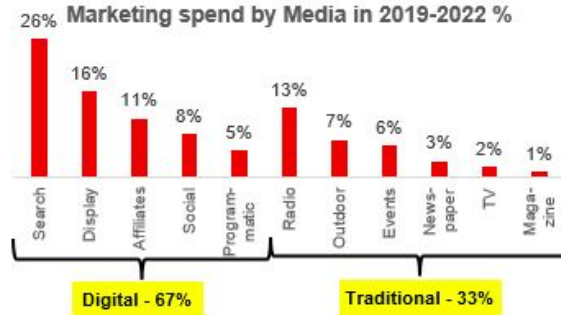


- Inflation rate
- Unemployment rate
- GDP
- USD/CAD exchange rate
- Population & households growth
- Household income
- Canada permanent & temporary residents
- Covid cases
- Canadian consumer seasonal behavioral periods (RRSP, March break, Back-to-school, Year-end-holiday)
- Google trends

* Data is necessary to explain baseline/BAU/non-marketing effects

3. Media Mix Model (MMM)

Data mined



3. Media Mix Model (MMM)

Addstock parameters for this exercise

- Addstock: accumulation of advertisements from previous weeks on this weeks NTB
- Decay: gradual decrease of accumulated adstock from previous weeks
- Lag: impact of marketing spend on NTB
- Saturation & diminishing effect: after specific point, incremental marketing spend leads to a smaller incremental NTB, and after a certain point to even zero incremental NTB (saturation)

Parameters from our champion Media Mix Model			
		Lags	<u>Addstock</u> decays
Traditional Media (longer lags, slower decays)	Radio	3	0.1
	Magazine	3	0.2
	Newspaper	3	0.5
	Outdoor	3	0.1
	TV	3	0.7
	Events	6	0.8
	Display	1	0.1
Digital Media (shorter lags, faster decays)	Programmatic	1	0.2
	Search	1	0.1
	Social	2	0.1
	Affiliates	1	0.1

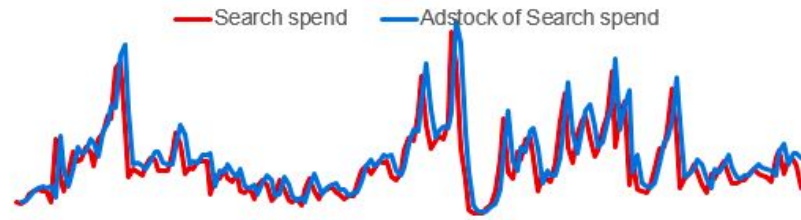
Spend TV from 3 week ago impact NTB (long lag)

Accumulated TV adstock decays 1.43x from last week (slow decay)

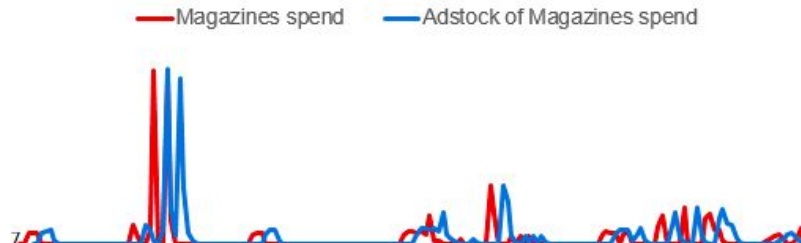
Spend on Search from 1 week ago impact NTB (short lag)

Accumulated Search adstock decays 10x from last week (fast decay)

Search: shorter lag and faster adstock decay



Magazines: longer lag and slower adstock decay

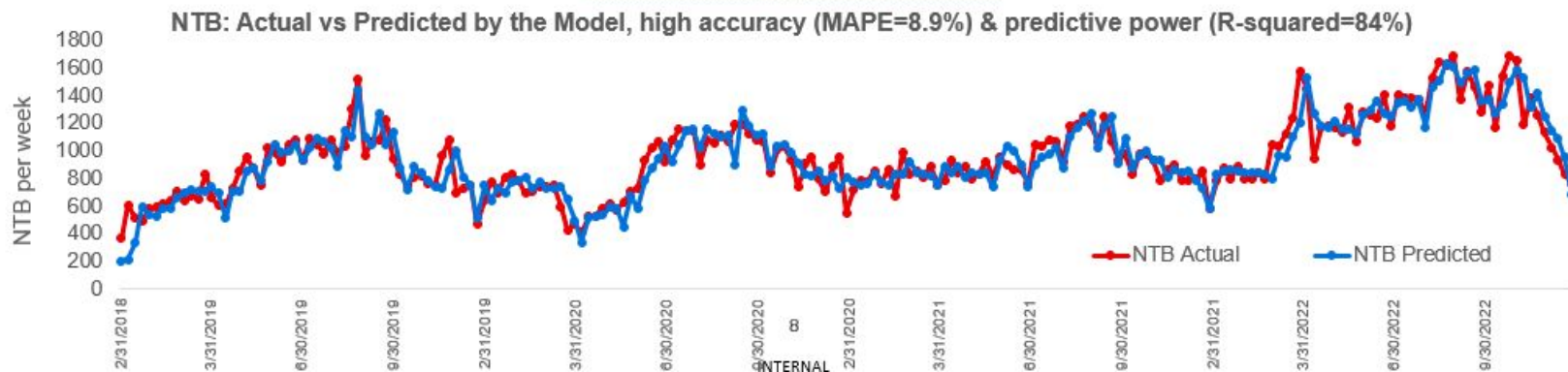


4. Results

Model Interpretation

$$NTB_t = \beta_1 Magazines_t + \beta_2 Search_t + \beta_3 Display_t + \beta_4 Affiliates_t + \beta_5 \#StatHolidays_t + \beta_6 EndOfYrHolidayPeriod_t + \beta_7 BackToSchoolPeriod_t + \beta_8 Unemployment_t + Seasonality_t + Cycle_t + \varepsilon_t$$

Model variable name	Regression coefficient	P-value (low p-value → high variable significance)
Marketing Spend on Magazines	0.0021	0.001
Marketing Spend on Search	0.0008	0.084
Marketing Spend on Display	0.0007	0.191
Marketing Spend on Affiliates	0.0009	0.073
Number of Statutory Holidays (in a week)	-134.6971	0.000
End of year holiday period	-56.3881	0.209
Back-to-school period	51.1732	0.181
Unemployment rate	1395.3004	0.000
Seasonality pattern	4200.0656	0.000
Cycle (cyclical patterns)	Multiple values for different parameters	0.000



4. Results

How results should look like if we hadn't used UCM

Estimated Equation

$$NTB_t = 3937.5328 + 0.00310 * AdStock(Paid Social_t, 0.9, lag 3) + 0.00311 * AdStock(Paid Search NTC_t, 0.9, lag 5) + 0.0170 * AdStock(Paid Search Others_t, 0.3, lag 3) + 284 * Seasonality_t + -26.2488 * Consumer Price Index_t + Error$$

variables	variable Description	Estimate	Std. Error	t value	Pr(> t)	vif
(Intercept)		3937.5328	679.1285566	5.79792	0.00	
Psocial_adst_dlag3_0.9	Adstock transformation of paid social spend for lag 3 and adstock rate 0.9	0.00310	0.000552917	5.60043	0.00	1.794091424
PS_NTC_adst_dlag5_0.9	Adstock transformation of paid search spend for NTC programs for lag and adstock rate 0.9	0.00311	0.001410627	2.20358	0.03	1.458319115
PS_Others_adst_dlag3_0.3	Adstock transformation of paid search spend for programs other than premier, mortgage, NTC, wealth for lag 3 and adstock rate 0.3	0.0170	0.0050393	3.36848	0.00	1.275218912
CPI	Consumer price index	-26.2488	5.323545067	-4.9307	0.00	1.492433644
seasonal_dummy	Seasonality index. =1 if the week has unusual spike in NTB i.e NTB values are far away from the mean NTB value, =0 if week has normal trend	284.4741	26.08084837	10.9074	< 2.22e-16	1.010677387
Multiple R-squared		55%				
Mape		11.8800				

5. Annex

Appendix

Variable reduction process insights

- To create a stable model, with stable variable coefficients, variables with high correlation to other variables have to be removed*
- Non-excluded but highly correlated variables do not increase model accuracy, but reduce model stability*

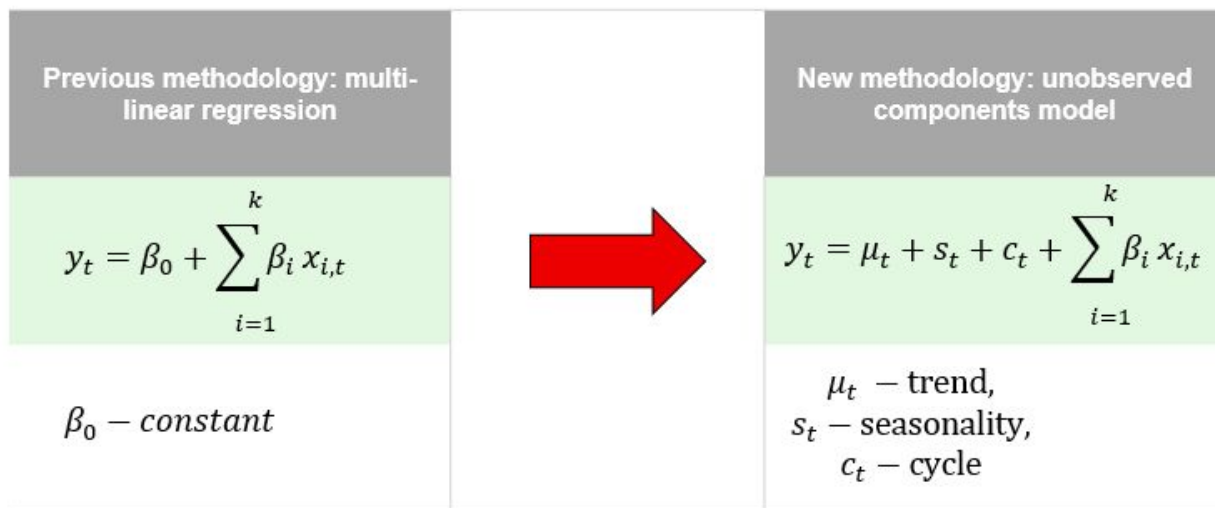
Variables excluded	Reason for exclusion
<u>Inflation rate</u>	High correlation with Unemployment rate, marketing channels and Admissions of Permanent Residents to Canada; high VIF
GDP	High correlation with Unemployment rate, USD/CAD rate, marketing channels and Admissions of Permanent Residents to Canada; high VIF
<u>USD/CAD exchange rate</u>	It was removed because it did not clearly establish a macroeconomic baseline on number of NTB activations; high VIF
Population growth	High correlation with Unemployment rate, marketing channels, Total households, Constant Dollar Household Average Income, and Admissions of Permanent Residents to Canada; high VIF
Households growth	High correlation with Unemployment rate, marketing channels, Total population, Constant Dollar Household Average Income, and Admissions of Permanent Residents to Canada; high VIF
<u>Household income</u>	High correlation with Unemployment rate, marketing channels, Total households, Total population, and Admissions of Permanent Residents to Canada; high VIF
Canadian consumer seasonal behavioral periods (Peak mortgage)	Counter intuitive impact on number of NTB activations

5. Annex

Appendix

Methodology enhancement

- Compared to the previous version of the Media Mix Model, mathematical methodology was modified from multi-linear regression to unobserved component model (UCM)
- High level, methodologies are similar, but UCM allows us to isolate the trend, seasonality and cycle components from the explanatory variables impacts. All these components are part of Baseline / BAU / Non-marketing effects. Otherwise the estimated coefficients would be spurious (false)



Gracias

